LAPORAN PRAKTIKUM MLOPS
STREAMLIT

Dosen pembimbing :

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Disusun oleh :

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SARJANA TERAPAN SAINS DATA TERAPAN

POLITEKNIK ELEKTRONIKA NEGRI SURABAYA

2024

Prepocessing :

```
import pandas as pd
from pycaret.datasets import get_data
from pycaret.classification import *

data = pd.read_csv('C:/Users/asuss/Downloads/streamlit/water_potability.csv')
data.fillna(data.mean(), inplace=True)
s = setup(data, target='Potability', session_id=123)
best = compare_models()
model = tune_model(best)
evaluate_model(model)
finalize_model(model)
save_model(model, 'water_pipeline')
```

	Description	Value
0	Session id	123
1	Target	Potability
2	Target type	Binary
3	Original data shape	(3276, 10)
4	Transformed data shape	(3276, 10)
5	Transformed train set shape	(2293, 10)
6	Transformed test set shape	(983, 10)
7	Numeric features	9
8	Preprocess	True
9	Imputation type	simple
10	Numeric imputation	mean
11	Categorical imputation	mode
12	Fold Generator	StratifiedKFold
13	Fold Number	10
14	CPU Jobs	-1
15	Use GPU	False
16	Log Experiment	False
17	Experiment Name	clf-default-name
18	USI	6f35

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
et	Extra Trees Classifier	0.6690	0.6727	0.3005	0.6673	0.4127	0.2275	0.2625	0.2770
qda	Quadratic Discriminant Analysis	0.6646	0.6733	0.3252	0.6366	0.4287	0.2267	0.2524	0.0550
rf	Random Forest Classifier	0.6563	0.6605	0.3140	0.6160	0.4142	0.2074	0.2310	0.8750
lightgbm	Light Gradient Boosting Machine	0.6415	0.6443	0.3889	0.5573	0.4565	0.2028	0.2102	0.1840
gbc	Gradient Boosting Classifier	0.6411	0.6088	0.2347	0.5989	0.3350	0.1522	0.1829	0.4770
nb	Naive Bayes	0.6145	0.5743	0.1866	0.5197	0.2738	0.0847	0.1059	0.0480
lda	Linear Discriminant Analysis	0.6123	0.5049	0.0067	0.5000	0.0132	0.0081	0.0448	0.0360
lr	Logistic Regression	0.6119	0.5060	0.0056	0.4000	0.0110	0.0068	0.0365	3.8620
ridge	Ridge Classifier	0.6119	0.5050	0.0056	0.4000	0.0110	0.0068	0.0365	0.0560
dummy	Dummy Classifier	0.6097	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0240
ada	Ada Boost Classifier	0.6018	0.5494	0.1789	0.4656	0.2568	0.0575	0.0680	0.5630
dt	Decision Tree Classifier	0.5765	0.5560	0.4626	0.4593	0.4597	0.1121	0.1125	0.0490
knn	K Neighbors Classifier	0.5556	0.5119	0.2972	0.4064	0.3425	0.0193	0.0201	0.0670
svm	SVM - Linear Kernel	0.5352	0.5077	0.3573	0.1982	0.2482	0.0063	0.0065	0.0390

Fold	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
0	0.6522	0.6593	0.4333	0.5735	0.4937	0.2365	0.2419
1	0.6087	0.6353	0.4444	0.5000	0.4706	0.1619	0.1627
2	0.6522	0.6333	0.3889	0.5833	0.4667	0.2236	0.2338
3	0.6157	0.6645	0.3667	0.5156	0.4286	0.1514	0.1563
4	0.6419	0.6628	0.5000	0.5488	0.5233	0.2375	0.2382
5	0.6507	0.6413	0.4045	0.5714	0.4737	0.2235	0.2310
6	0.6594	0.6462	0.4607	0.5775	0.5125	0.2558	0.2597
7	0.6943	0.6983	0.5169	0.6301	0.5679	0.3350	0.3389
8	0.6769	0.6738	0.4494	0.6154	0.5195	0.2849	0.2928
9	0.6681	0.6861	0.4719	0.5915	0.5250	0.2749	0.2790
Mean	0.6520	0.6601	0.4437	0.5707	0.4981	0.2385	0.2434
Std	0.0245	0.0206	0.0450	0.0383	0.0375	0.0519	0.0524

```
Fitting 10 folds for each of 10 candidates, totalling 100 fits
Original model was better than the tuned model, hence it will be returned. NOTE: The display metrics are for the tuned model (not the original one).

Transformation Pipeline and Model Successfully Saved

(Pipeline(memory=Memory(location=None),
          steps=[('numerical_imputer',
                  TransformerWrapper(exclude=None,
                                     include=['ph', 'Hardness', 'Solids',
                                              'Chloramines', 'Sulfate',
                                              'Conductivity', 'Organic_carbon',
                                              'Trihalomethanes', 'Turbidity'],
                                     transformer=SimpleImputer(add_indicator=False,
                                                               copy=True,
                                                               fill_value=None,
                                                               keep_empty_features=False,
                                                               missing_values=nan,
                                                               strategy='mean...
                  ExtraTreesClassifier(bootstrap=False, ccp_alpha=0.0,
                                       class_weight=None, criterion='gini',
                                       max_depth=None, max_features='sqrt',
                                       max_leaf_nodes=None, max_samples=None,
                                       min_impurity_decrease=0.0,
                                       min_samples_leaf=1, min_samples_split=2,
                                       min_weight_fraction_leaf=0.0,
                                       monotonic_cst=None, n_estimators=100,
                                       n_jobs=-1, oob_score=False,
                                       random_state=123, verbose=0,
                                       warm_start=False))],
          verbose=False),
 'water_pipeline.pkl')
```

webApps.py :

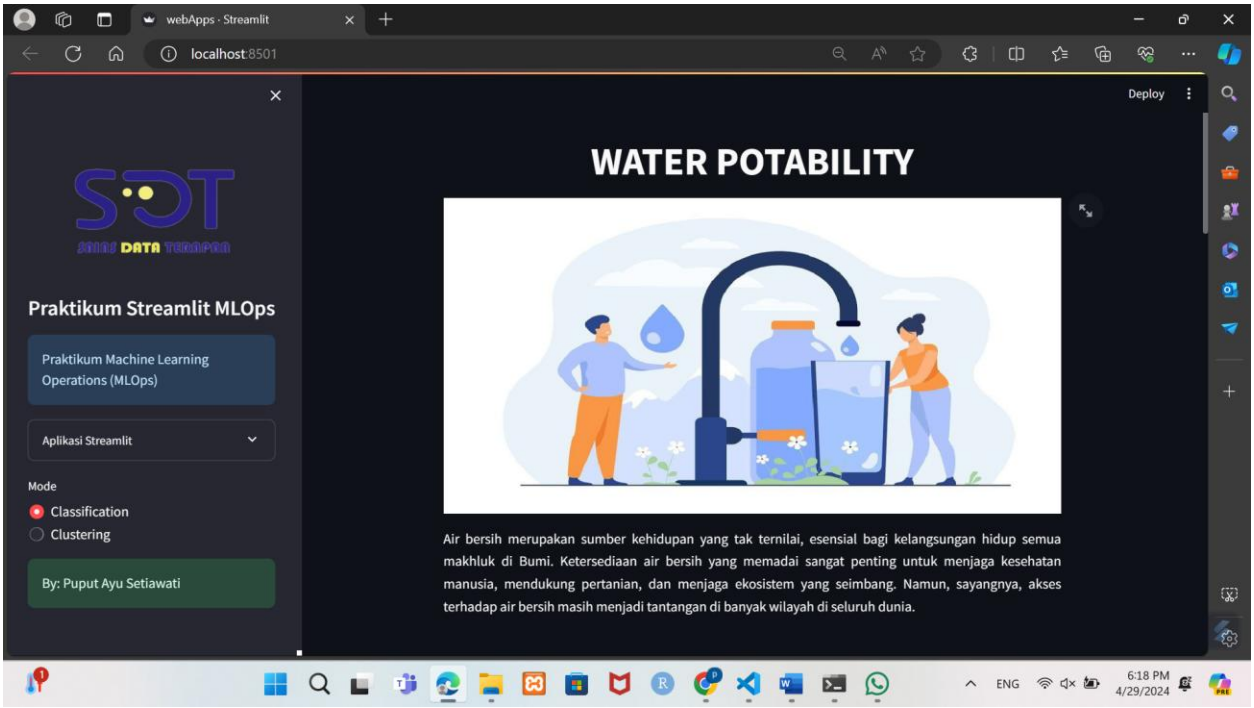
```
import pandas as pd
import seaborn as sns
import streamlit as st
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
from pycaret.classification import load_model, predict_model
from pycaret.clustering import setup, create_model, assign_model

# Load model classification
classification_model = load_model('water_pipeline')

# Menentukan fungsi classification yang akan dipanggil
def predict_classification(model, input_df):
    predictions_df = predict_model(classification_model, data=input_df)
    st.write(predictions_df)
    prediction = predictions_df.iloc[0, 0]
    return prediction

# Menentukan fungsi clustering yang akan dipanggil
def perform_clustering(data, num_clusters, method):
    clustering_model = create_model(method, num_clusters=num_clusters)
    assigned_clusters = assign_model(clustering_model)
    return assigned_clusters

def display_classification_interface():
    st.title("Classification Water Potability")

    ph = st.number_input('pH (0-14)', min_value=0.0, max_value=14.0, value=7.0, step=0.1)
    hardness = st.slider('Hardness (ppm)', min_value=0, max_value=500, value=100)
    solids = st.slider('Total Dissolved Solids (ppm)', min_value=0, max_value=1000,
value=200)
    chloramines = st.number_input('Chloramines (ppm)', min_value=0.0, max_value=15.0,
value=10.0, step=0.1)
    sulfate = st.slider('Sulfate (ppm)', min_value=0, max_value=500, value=100)
    conductivity = st.slider('Conductivity (µS/cm)', min_value=0, max_value=2000, value=500)
    organic_carbon = st.number_input('Organic Carbon (ppm)', min_value=0, max_value=50,
value=10)
    trihalomethanes = st.slider('Trihalomethanes (ppm)', min_value=0, max_value=200,
value=50)
    turbidity = st.number_input('Turbidity (NTU)', min_value=0, max_value=10, value=5)

    input_dict = {
        'ph': ph,
        'Hardness': hardness,
        'Solids': solids,
        'Chloramines': chloramines,
        'Sulfate': sulfate,
        'Conductivity': conductivity,
        'Organic_carbon': organic_carbon,
        'Trihalomethanes': trihalomethanes,
        'Turbidity': turbidity
```

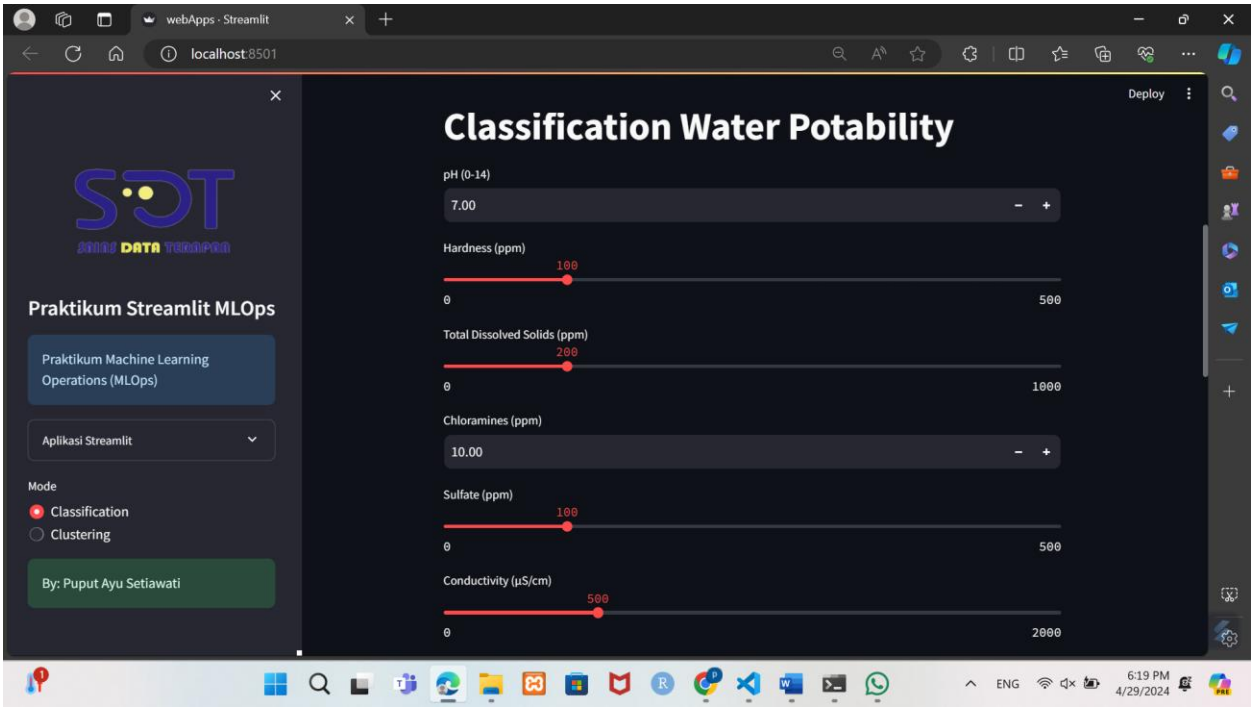
```
    }

    input_df = pd.DataFrame([input_dict])

    if st.button("Predict"):
        output = predict_classification(model=classification_model, input_df=input_df)
        if output:
            st.success('Air Layak Diminum')
        else:
            st.error('Air Tidak Layak Diminum')


        # Visualisasi perbandingan data atribut classification
        st.subheader("Comparison Atribut")
        plot_data = input_df.iloc[0]
        plot_data.index = ['pH', 'Hardness', 'Solids', 'Chloramines', 'Sulfate',
'Conductivity', 'Organic Carbon', 'Trihalomethanes', 'Turbidity']
        fig, ax = plt.subplots(figsize=(10, 6))
        sns.barplot(x=plot_data.index, y=plot_data.values)
        plt.xticks(rotation=45)
        plt.xlabel("Atribut")
        plt.ylabel("Count")
        plt.title("Comparison Data Atribut")
        st.pyplot(fig)

def display_clustering_interface():
    st.title("Clustering Water Potability")

    num_clusters = st.number_input("Input the number of clusters", min_value=2,
max_value=10, value=3)
    st.write("Running KMeans clustering with the parameters:")
    st.write("Number of clusters:", num_clusters)

    if st.button("Clustering"):
        # Load model clustering
        data_clustering =
pd.read_csv('C:/Users/asuss/Downloads/streamlit/water_potability.csv')
        data_clustering.fillna(data_clustering.mean(), inplace=True)

        clustering_setup = setup(data=data_clustering, session_id=123)
        clustering_model = create_model('kmeans', num_clusters=num_clusters)
        assigned_clusters = assign_model(clustering_model)

        st.write("Data clustering results:")
        st.write(assigned_clusters.head(20))

        # Visualisasi hasil clustering menggunakan PCA
        st.subheader("Clustering Results With PCA")
        pca = PCA(n_components=2)
        pca_result = pca.fit_transform(assigned_clusters.iloc[:, :-1])
        pca_df = pd.DataFrame(data=pca_result, columns=['PC1', 'PC2'])
        pca_df['Cluster'] = assigned_clusters['Cluster']
```

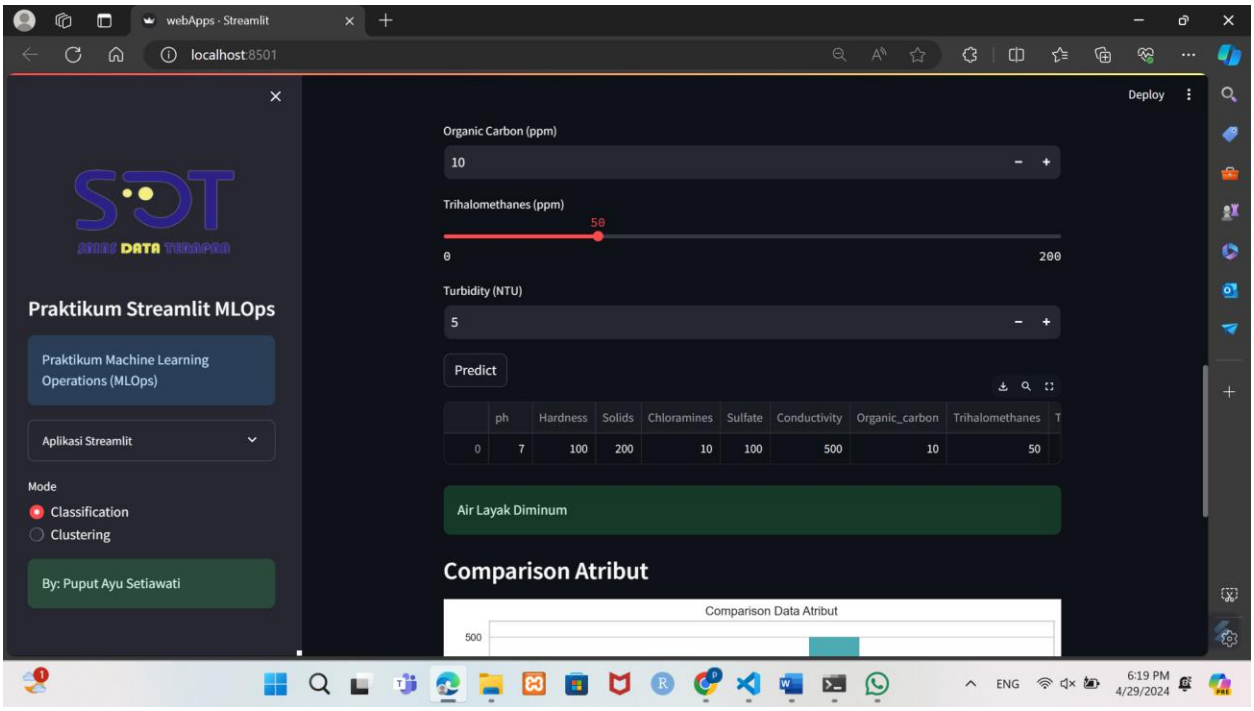
```
        plt.figure(figsize=(10, 6))
        sns.scatterplot(data=pca_df, x='PC1', y='PC2', hue='Cluster', palette='deep',
legend='full')
        plt.title('Clustering Results (PCA)')
        plt.xlabel('Principal Component 1')
        plt.ylabel('Principal Component 2')
        st.pyplot()

def run():
    from PIL import Image
    image = Image.open('C:/Users/asuss/Downloads/streamlit/image/logo_sdt.png')
    image_air = Image.open('C:/Users/asuss/Downloads/streamlit/image/air.jpg')

    # Add sidebar
    st.sidebar.image(image)
    st.sidebar.title('Praktikum Streamlit MLOps')
    st.sidebar.info("Praktikum Machine Learning Operations (MLOps)")
    with st.sidebar.expander("Aplikasi Streamlit"):
        st.markdown("""Aplikasi streamlit ini melakukan classification dan clustering water
potability yang berfungsi untuk memprediksi apakah air dari sumber tertentu layak untuk
dikonsumsi atau tidak berdasarkan beberapa fitur yang diukur.
        Klasifikasi bertujuan untuk mengidentifikasi air yang memenuhi standar potability,
sementara pengelompokan (clustering) membantu mengidentifikasi pola-pola alami dalam data
yang mungkin tidak terlihat secara langsung,
        seperti kemungkinan cluster air yang memiliki karakteristik serupa yang tidak
memenuhi standar potability. Dengan menggunakan metode ini, aplikasi dapat membantu konsumen
untuk mengambil keputusan yang lebih baik terkait penggunaan air.""")

    # Add title dan keterangan di main interface
    st.markdown("<h1 style='text-align: center;'>WATER POTABILITY</h1>",
unsafe_allow_html=True)
    st.image(image_air)
    st.markdown("<p style='text-align: justify;'>Air bersih merupakan sumber kehidupan yang
tak ternilai, esensial bagi kelangsungan hidup semua makhluk di Bumi. Ketersediaan air
bersih yang memadai sangat penting untuk menjaga kesehatan manusia, mendukung pertanian, dan
menjaga ekosistem yang seimbang. Namun, sayangnya, akses terhadap air bersih masih menjadi
tantangan di banyak wilayah di seluruh dunia.</p>", unsafe_allow_html=True)

    # Add button untuk select mode
    mode = st.sidebar.radio("Mode", ["Classification", "Clustering"])
    st.sidebar.success("By: Puput Ayu Setiawati")

    if mode == "Classification":
        display_classification_interface()
    elif mode == "Clustering":
        display_clustering_interface()

if __name__ == '__main__':
    run()
```

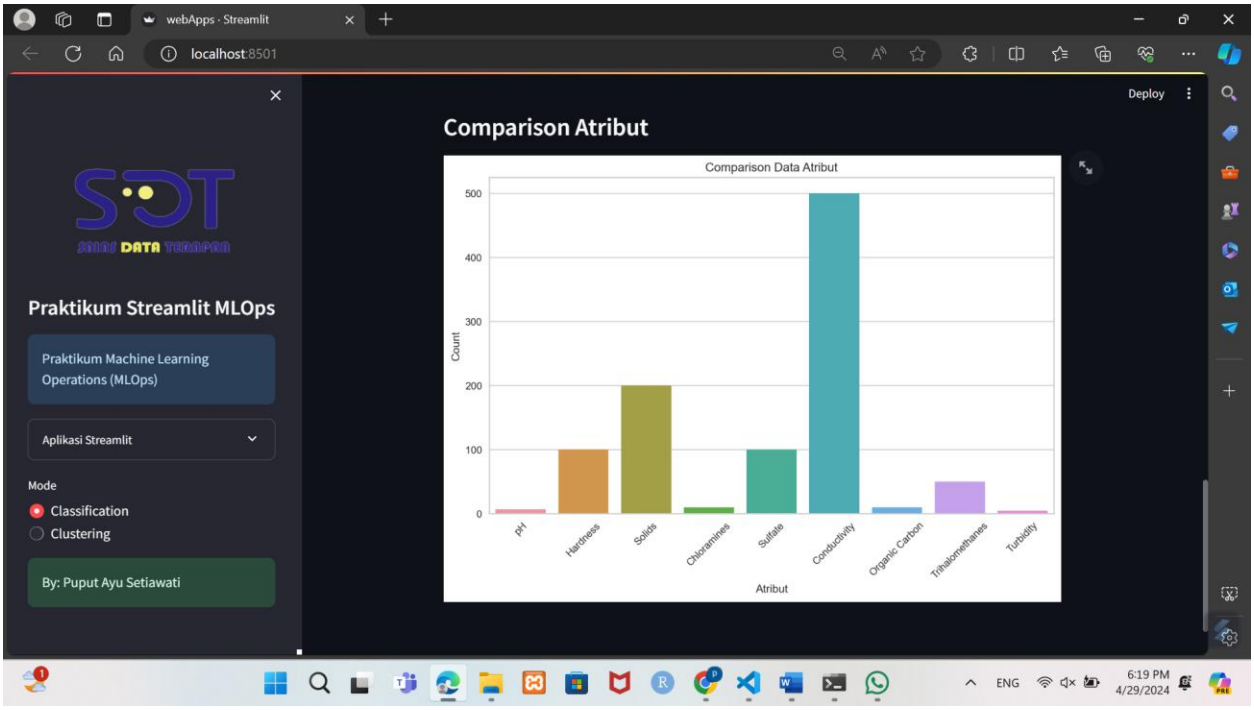
Aplikasi Website :

Mode Classification

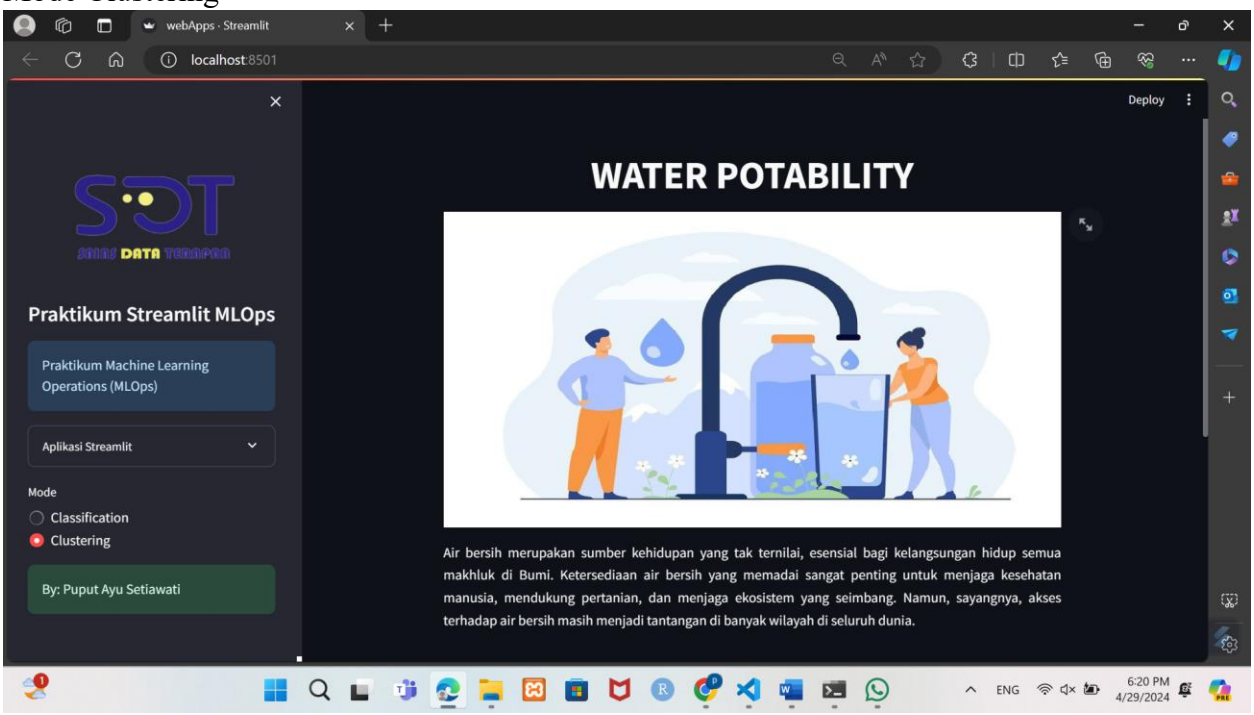


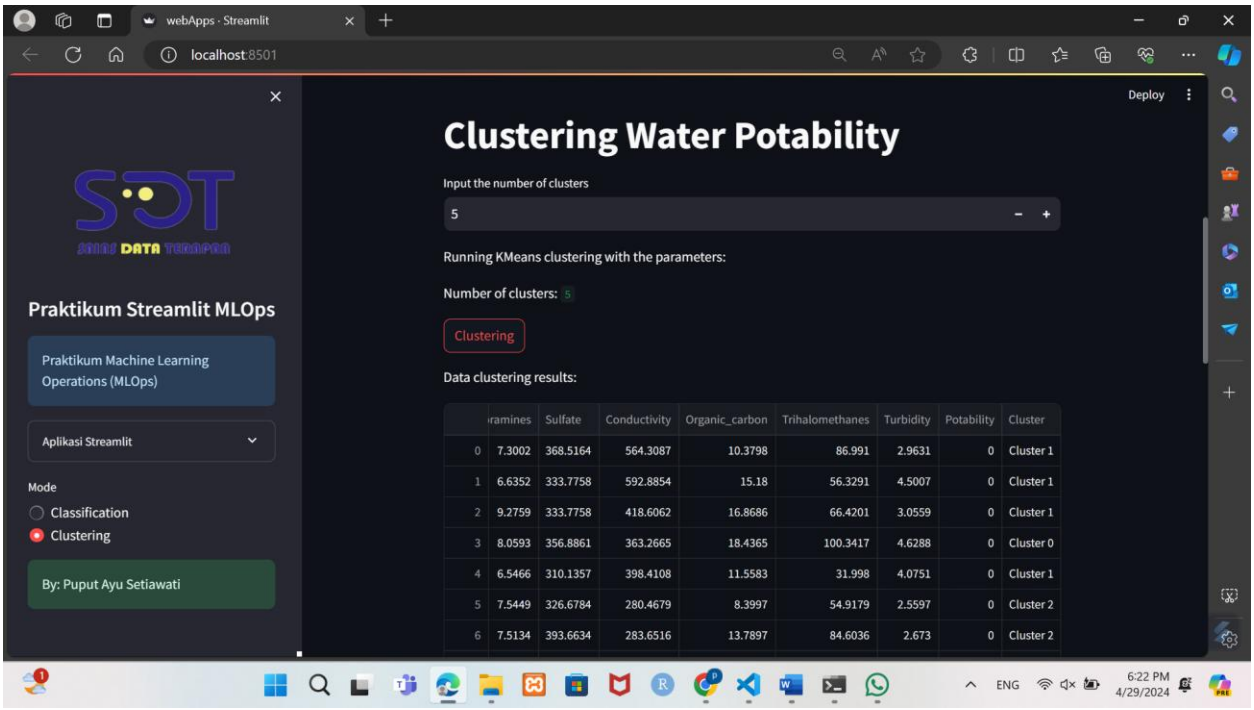


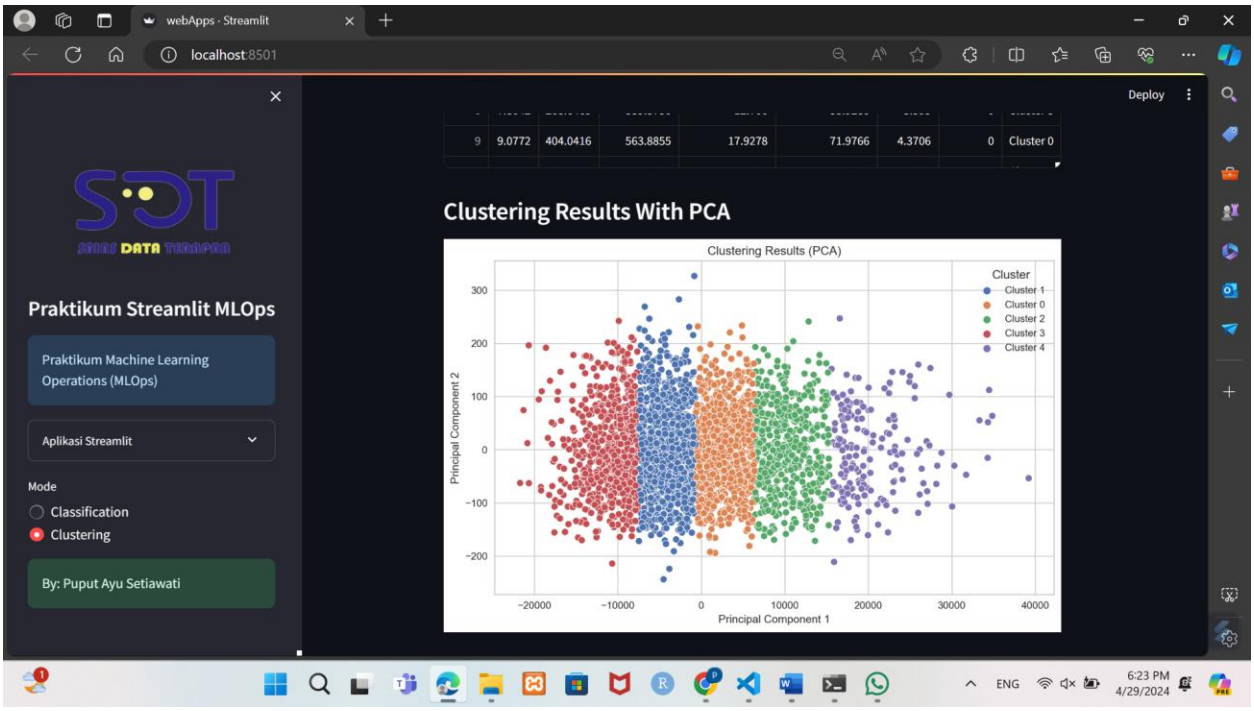

webApps · Streamlit
localhost:8501
Deploy
Praktikum Streamlit MLOps
Praktikum Machine Learning Operations (MLOps)
Aplikasi Streamlit
Mode
Classification
Clustering
By: Puput Ayu Setiawati
Organic Carbon (ppm)
10
Trihalomethanes (ppm)
50
0
200
Turbidity (NTU)
5
Predict
ph Hardness Solids Chloramines Sulfate Conductivity Organic_carbon Trihalomethanes
0 7 100 200 10 100 500 10 50
Air Layak Diminum
Comparison Atribut
Comparison Data Atribut
500
ENG
6:19 PM
4/29/2024


webApps · Streamlit
localhost:8501
Deploy
Praktikum Streamlit MLOps
Praktikum Machine Learning Operations (MLOps)
Aplikasi Streamlit
Mode
Classification
Clustering
By: Puput Ayu Setiawati
Comparison Atribut
Comparison Data Atribut
Count
500
400
300
200
100
0
pH
Hardness
Solids
Chloramines
Sulfate
Conductivity
Organic Carbon
Trihalomethanes
Turbidity
Atribut
ENG
6:19 PM
4/29/2024

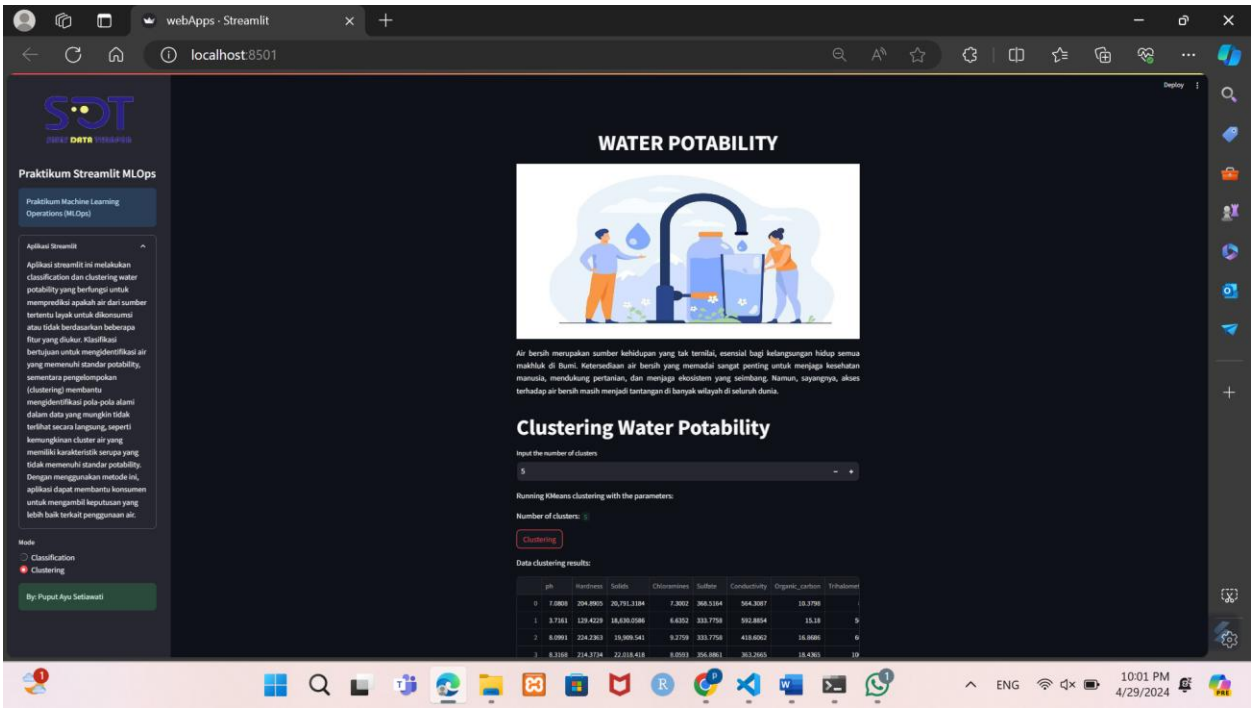
Mode Clustering






webApps - Streamlit
localhost:8501
Deploy
Praktikum Streamlit MLOps
Praktikum Machine Learning Operations (MLOps)
Aplikasi Streamlit
Mode
Classification
Clustering
By: Puput Ayu Setiawati
Clustering Results With PCA
Clustering Results (PCA)
Cluster
Cluster 1
Cluster 0
Cluster 2
Cluster 3
Cluster 4
Principal Component 2
Principal Component 1
ENG
6:23 PM
4/29/2024

Sidebar


webApps - Streamlit
localhost:8501
Praktikum Streamlit MLOps
WATER POTABILITY
Clustering Water Potability
ENG
10:01 PM
4/29/2024